

PE-EC801B — Fiber Optic Communication

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Module 2 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

Contents

1	Signal Attenuation in Optical Fibers	2
1.1	Power Loss Formula	2
1.2	Causes of Attenuation	2
2	Dispersion in Optical Fibers	3
2.1	Types of Dispersion	3
3	Fiber Fabrication Processes	4
3.1	MCVD — Modified Chemical Vapor Deposition	4
3.2	OVD — Outside Vapor Deposition	4
3.3	VAD — Vapor Phase Axial Deposition	4
3.4	PCVD — Plasma Chemical Vapor Deposition	4
4	Fiber Measurement Techniques	5
4.1	OTDR — Principle and Trace Analysis	5
4.2	Cutback Method — Attenuation Measurement	6
4.3	Interferometric Method — Dispersion Measurement	6

1. Signal Attenuation in Optical Fibers

Definition of Attenuation

Attenuation (or fiber loss) is the reduction in optical power as light propagates along a fiber. It is expressed in dB/km and is a primary factor limiting the repeater spacing in a fiber optic link.

1.1 Power Loss Formula

If P_{in} is the optical power launched into a fiber of length L and P_{out} is the power at the far end:

Attenuation Coefficient

$$\alpha \text{ (dB/km)} = \frac{10}{L} \log_{10} \left(\frac{P_{in}}{P_{out}} \right)$$

$$P_{out} = P_{in} \cdot 10^{-\alpha L/10}$$

At 1310 nm: $\alpha \approx 0.35$ dB/km; at 1550 nm: $\alpha \approx 0.20$ dB/km (for standard SMF).

1.2 Causes of Attenuation

1. Absorption Losses:

- **Intrinsic absorption:** Due to electronic resonances (UV) and vibrational resonances (IR) of the silica (SiO_2) matrix. This sets the fundamental lower limit on attenuation.
- **Extrinsic absorption:** Caused by **OH^- ions** (hydroxyl impurities from water vapor during fabrication) — produces absorption peaks at 1.38 μm , 950 nm, and 725 nm. Also caused by metal ion impurities (Fe, Cu, Mn).

2. Scattering Losses:

- **Rayleigh scattering:** Caused by random microscopic density and composition fluctuations (dimensions $\ll \lambda$) frozen in during fiber fabrication. Varies as λ^{-4} — dominant at short wavelengths. The Rayleigh scattering coefficient $A_R \approx 0.8 \text{ dB/km} \cdot \mu\text{m}^4$.
- **Mie scattering:** Due to inhomogeneities comparable in size to the wavelength (imperfect core-cladding interfaces, diameter fluctuations, bubbles). Usually small in modern fibers.

3. Bending Losses:

- **Macrobending:** When the fiber is bent to a radius $R < R_c$ (critical radius), the evanescent field at the outer edge must travel faster than the speed of light, so energy radiates out.
- **Microbending:** Caused by small random deformations of the fiber axis due to lateral stress during cabling. Causes mode coupling and energy transfer to radiation modes.

2. Dispersion in Optical Fibers

Dispersion — Definition

Dispersion is the spreading of optical pulses as they propagate along a fiber, caused by different frequency components or different modes traveling at different speeds. It limits the bandwidth and hence the data rate of the fiber link.

2.1 Types of Dispersion

Intermodal (Modal) Dispersion: Occurs in multimode fibers because different modes travel at different velocities. The fastest mode (along the axis) and the slowest mode differ in travel time. For MMSI:

$$\delta\tau_{\text{modal}} = \frac{n_1 L}{c} \cdot \Delta = \frac{n_1 L}{c} \cdot \frac{n_1 - n_2}{n_1}$$

Modal dispersion is **absent in single-mode fibers** and greatly reduced in MMGI fibers.

Intramodal (Chromatic) Dispersion: Occurs within a single mode and has two components:

- 1. Material dispersion:** The refractive index $n(\lambda)$ varies with wavelength. Different spectral components of the source pulse travel at different speeds. Characterized by material dispersion parameter D_m (ps/nm·km). Zero at $\lambda_0 \approx 1.27 \mu\text{m}$ for silica.
- 2. Waveguide dispersion:** Even for a fixed wavelength, the propagation constant β depends on the ratio $V = 2\pi a \text{NA}/\lambda$, so the group velocity varies with wavelength due to the fiber geometry. Significant in SMF. Characterized by D_w (ps/nm·km). Negative at 1550 nm.
- 3. Chromatic (total) dispersion:** $D = D_m + D_w$ (ps/nm·km). For standard SMF: $D = 0$ at $\lambda \approx 1310 \text{ nm}$; $D \approx +17 \text{ ps/nm·km}$ at 1550 nm.

Dispersion Formulas

RMS pulse broadening
(chromatic)

$$\sigma = D \cdot L \cdot \Delta\lambda \quad (\text{ps})$$

Bandwidth–length
product

$$B \cdot L \leq \frac{0.44}{\sigma/L} \quad (\text{Gaussian pulse})$$

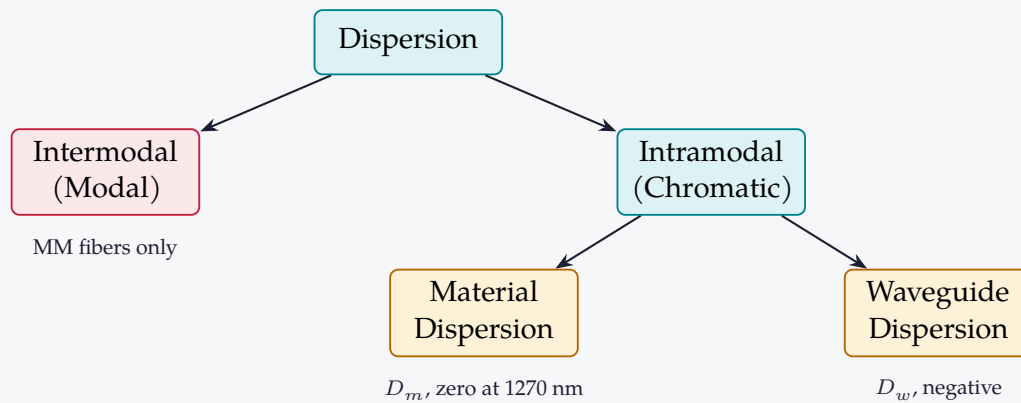
Modal dispersion
(MMSI)

$$\delta\tau = \frac{n_1 L \Delta}{c}$$

Modal dispersion
(MMGI, $\alpha=2$)

$$\delta\tau \approx \frac{n_1 L \Delta^2}{2c}$$

Dispersion Types — Classification Chart



3. Fiber Fabrication Processes

Overview of Fabrication Methods

Modern silica optical fibers are made by **Chemical Vapor Deposition (CVD)** processes, where dopant-laden gases react at high temperature to deposit glass layers. The four main processes are MCVD, OVD, VAD, and PCVD.

3.1 MCVD — Modified Chemical Vapor Deposition

Dopant gases (SiCl_4 , GeCl_4 , POCl_3 , BCl_3) are flowed inside a silica tube. An external oxygen–hydrogen flame traverses the tube, causing oxidation and deposition of glass soot inside. Multiple passes deposit core and cladding layers. The tube is then collapsed into a solid **preform** by heating to $> 2000^\circ\text{C}$.

3.2 OVD — Outside Vapor Deposition

Soot is deposited on the **outside** of a bait rod (mandrel) via a flame hydrolysis burner. The rod is removed and the porous soot boule is consolidated (sintered) in a furnace to form a clear glass preform, then drawn into fiber.

3.3 VAD — Vapor Phase Axial Deposition

Soot is deposited axially at the end of a rotating preform billet using burners. The porous preform grows **upward** and is continuously pulled into a consolidation furnace. Allows continuous fiber production. Produces very low OH^- content.

3.4 PCVD — Plasma Chemical Vapor Deposition

Uses a non-resonant microwave plasma (instead of a flame) inside a silica tube to drive the CVD reaction. Very precise compositional control; used for special fibers.

Feature	MCVD	OVD	VAD	PCVD
Deposition location	Inside tube	Outside rod	Axial end	Inside tube
Heat source	External flame	Flame hydrolysis	Flame	Microwave plasma
Collapse needed?	Yes	Yes (soot)	Yes	Yes
OH ⁻ impurity	Moderate	Low	Very low	Low
Production rate	Medium	High	High (continuous)	Low
Profile control	Good	Good	Good	Excellent

4. Fiber Measurement Techniques

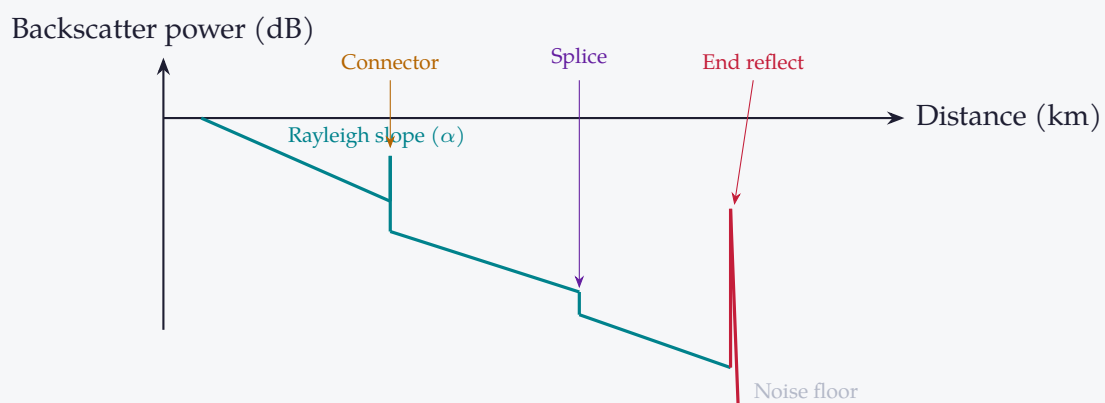
Optical Time-Domain Reflectometer (OTDR)

An OTDR is a sophisticated instrument that characterizes a fiber by injecting a short optical pulse and measuring the Rayleigh backscattered and Fresnel-reflected light as a function of time (and hence distance). It gives a **complete loss profile** of the fiber from a single end.

4.1 OTDR — Principle and Trace Analysis

The OTDR launches a pulse into the fiber. Light is continuously backscattered toward the OTDR. Splice losses, connector losses, and faults appear as steps or spikes. The time delay t between launch and detection corresponds to a distance $d = ct/(2n_1)$.

Typical OTDR Trace



Key OTDR terms:

- **Dead zone:** Distance near the launch connector where the receiver is saturated after a large Fresnel reflection — events within this zone cannot be detected. Typical: 1–10 m.

- **Dynamic range:** The difference between the initial backscatter power level and the noise floor (dB). Larger dynamic range $\Rightarrow$ longer fiber can be characterized.

4.2 Cutback Method — Attenuation Measurement

The most accurate method for measuring attenuation. Launch power P_1 is measured by cutting the fiber close to the input (~ 2 m from the source). Then the full-length measurement P_2 is taken. No need to disturb the launch conditions.

$$\alpha = \frac{10}{L} \log_{10} \left(\frac{P_1}{P_2} \right) \quad \text{dB/km}$$

4.3 Interferometric Method — Dispersion Measurement

Used to measure **group delay** vs. wavelength. A broadband or tunable source illuminates the fiber; the output is compared against a reference path in a Mach-Zehnder or Michelson interferometer. Phase shift vs. wavelength gives the group delay $\tau(\lambda)$, from which $D(\lambda) = (1/L) d\tau/d\lambda$ (ps/nm·km).

Quick Revision — Module 2

Signal Degradation, Fabrication, and Measurement | PE-EC801B

Key Formulas at a Glance

Attenuation coefficient	$\alpha = \frac{10}{L} \log_{10}(P_{in}/P_{out}) \quad (\text{dB/km})$
Output power	$P_{out} = P_{in} \cdot 10^{-\alpha L/10}$
Rayleigh scattering	Varies as λ^{-4}
RMS pulse spread	$\sigma = D \cdot L \cdot \Delta\lambda$
Modal dispersion (MMSI)	$\delta\tau = n_1 L \Delta/c$
Modal dispersion (MMGI)	$\delta\tau \approx n_1 L \Delta^2/(2c)$
Chromatic dispersion	$D = D_m + D_w \quad (\text{ps/nm} \cdot \text{km})$
BW-length product	$B \cdot L \leq 0.44/(\sigma/L)$
OTDR distance	$d = ct/(2n_1)$

One-Line Definitions

- **Attenuation:** Reduction in optical power along the fiber; expressed in dB/km.
- **Rayleigh scattering:** Scattering from density fluctuations; $\propto \lambda^{-4}$; dominant at short λ .
- **Modal dispersion:** Pulse broadening from different modes arriving at different times; absent in SMF.
- **Material dispersion:** Pulse broadening due to $n(\lambda)$ variation; zero at 1270 nm in silica.
- **Waveguide dispersion:** Pulse broadening due to wavelength-dependent propagation constant.
- **OTDR:** Instrument that measures fiber loss profile from a single end using backscattered light.
- **Dead zone:** Region near the OTDR launch where saturation prevents event detection.
- **Cutback method:** Gold-standard destructive technique for measuring fiber attenuation accurately.
- **MCVD:** Modified CVD — glass deposited inside a silica tube by an external traversing flame.

Frequently Asked / Viva Questions

Q1. What are the main causes of attenuation in an optical fiber?

Absorption (intrinsic UV/IR resonances and extrinsic OH^- impurities), Rayleigh scattering (λ^{-4}), Mie scattering, macrobending, and microbending losses.

Q2. Why is 1310 nm and 1550 nm preferred as operating windows?

At 1310 nm, chromatic dispersion is near zero for SMF. At 1550 nm, Rayleigh scattering is low, giving minimum attenuation (~ 0.2 dB/km) for standard silica SMF.

Q3. What is Rayleigh scattering and why is it important?

It results from random sub-wavelength density fluctuations in silica. Since it varies as λ^{-4} , it is the dominant loss at short wavelengths and sets the lower limit for SMF attenuation.

Q4. Distinguish between intermodal and intramodal dispersion.

Intermodal dispersion occurs in multimode fibers when different modes travel at different velocities. Intramodal (chromatic) dispersion occurs within a single mode due to material and waveguide effects.

Q5. Why does MMGI have lower modal dispersion than MMSI?

The parabolic index profile of MMGI makes higher-order modes travel faster (in lower-index outer regions), nearly equaling the speed of lower-order modes. This self-focusing effect reduces differential mode delay.

Q6. What is bandwidth–length product? Give a typical value.

It is the product $B \times L$ (MHz·km or GHz·km), measuring the information capacity of a fiber per unit length. Typical: MMSI ~ 20 MHz·km; MMGI ~ 1000 MHz·km; SMF > 100 GHz·km.

Q7. Explain the OTDR principle briefly.

A short optical pulse is launched into the fiber. The OTDR measures the time-resolved backscattered signal. Distance is calculated as $d = ct/(2n)$. The trace shows attenuation slope, splices, connectors, and fiber end.

Q8. What is the dead zone in an OTDR?

The dead zone is the distance near the OTDR launch connector where the receiver is saturated by the large Fresnel reflection, making it unable to detect nearby events. Typically 1–10 m.

Q9. Describe the cutback method for attenuation measurement.

Launch power P_1 is measured by cutting the fiber ~ 2 m from the source. Then full-length output P_2 is measured. $\alpha = (10/L) \log(P_1/P_2)$. Most accurate but destructive.

Q10. What is the difference between macrobending and microbending?

Macrobending is a large-radius bend that causes the evanescent field to radiate. Microbending consists of random small-amplitude axis deformations from cabling stress, causing mode coupling and loss.

Frequently Asked / Viva Questions (contd.)

Q11. What is MCVD? How is the preform made?

Modified CVD. Dopant gases flow inside a rotating silica tube; an external oxy-hydrogen flame deposits glass soot on the inner wall. Multiple passes build up core/cladding layers, then the tube is collapsed into a solid preform.

Comparison: Fabrication Methods

Method	Deposition	Heat Source	OH ⁻	Rate
MCVD	Inside tube	External flame	Moderate	Medium
OVD	Outside rod	Flame hydrolysis	Low	High
VAD	Axial end	Flame	Very low	High (cont.)
PCVD	Inside tube	Microwave plasma	Low	Low